

LLM 2-04 4-Days Report..

Data Engineering - Pipeline yang

fund.

Eng awal LLM Domaini yonohi,
Janitor olindad

Data set osasan sugeng face
dan olindad

Repo struktura yarotib olindad,

[Repo structure]

my-llm /

data / raw / cleaned / filtered / final /

scripts / prepare-dataset build-dataset

loader / (5-days)

dataset / (1-days)

model / (8-days)

training / (10-days)

evaluation / (11-days)

deployment / (8-days)

configs / data.yaml run-ot.yaml

MODEL_CARD.md

README.md

.gitignore

Data/Dataset-Card.md

2. Q & A

1. good: Data re checkpoint Git ga
hech gachon tushmaydi - fagat

olarni qayta yaratadigan leod
tushadi (git → server → remote → git history)

2-qildas har bir run ox config faqat
ega - tushirib xolatlarini taqqoslash
uchun qayta tashlash uchun yaratilgan

2-qildas dataset TinyStories hichok
xalqolar dataseti manba: hugging
face

Data download qilinganda reponi
ichiga tushmaydi C olishga tushadi

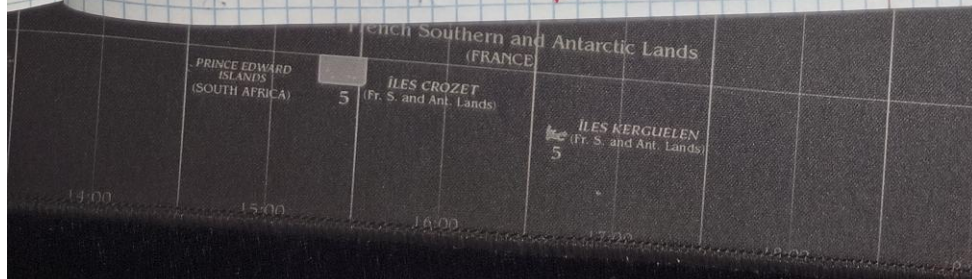
import hashlib - hash u-x sarfollarni
incollash u-x hash

!pip install -q datasets

Bu orqaga hugging Face dan datani
yuklab olishga ruxsat beradi
bu yerda (-q) bu faqat sawiq
masalarni oling hamma boshqa
dependencieslarni usun qilib kopya
yolib yubarmagan olingan

cleaning

raw HTML (loadi)
→ taglarni aloqalash
→ unicode normalizatsiya
→ whitespace qirish
→ nav/footer/cookie banner/ reklama
→ navigatsiya tusatosh
→ bosh yoki bursolgan hujjat → drop
→ final document



Cleaning *bozqochiluv*

HTML / markup - tag, script, style, navigat

Unicode NFKC - for xol belgi for xil koynatish

White space - ketma-ket boshlov, ya - bosh qatorlarni silqish

Mojibaze - noto'g'ri encoding natijasi (A©, & € -) tusatib berish

Control characters - ko'rinmas belgilar olib tashlanadi

Drop - ~~boz~~ lasilgan strukturalar, buzilgan hujjatlar.

Demak bizda 2 mln dan ortiq input bor va foydalanishdan qo'ng'ir qoznatilgan soni 250 taga kamayib qabul.

Filtering

Filtering signallari

Language ID - confidence threshold bilan

Length bounds - juda qisqa va juda uzun hujjatlar

Mean word length - anomaly qoznat - oqlat

Alphabetic ratio - belgilar va raqamlar ulushi

Punctuation ending ratio - nechta qator muqata bilan tugaydi

Repeated-line ratio - hujjat ichidagi takror

Repeated n-gram ratio - shablon va SEO spam belgisi

Block list - ma'lum spam domrlarini
reza pattenlar

Threshold bilan 2 xil xolat

Diffus threshold - kerakli hujjat
tushib qoladi: dialekt, kam uch-
raydigan, domen, qisqa lekin
qimmatli matn.

Yunchov threshold - shovqin, takror
to'sik matn qoladi.

Yechim Muhammad threshold emas
O'zgarigan threshold.

Deduplication

Takroriy qoymatlarni tozalash.
Bireal hashlarni song duplication

Near-duplicate

Shingles - hujjat ketma-ket n-gram
toplamga aylantiriladi.

Min-hash - har topdam kichik insoga
soyiladi.

LSH - o'xshash insolar bir bucket
ga tushadi, faqat ular solish.

Matnlar barcha juftlikni solishor
masalan o'xshashlarni topish.

PII Scrubbing

Xaief boydcha garion turli
xaiflarni oldini olish uchun
maska bilan almashtiriladi
3xdl turli bor

Redaction - Completely removes

Anonymization - generalizes

Pseudonymization - replaces with
fake data

Magsadi **Secure Dataset**

LLM qurganda Data report yoziladi
ichida quyidagilar bo'lishi shart

Raw - hujjat soni, belgi soni, haqiqi (MB)

Cleaning - kvado/chozoli, drop sabablari

Filtering - har bor boydcha soni
uchun alohida drop soni

Deduplication - exact dupli-n soni, near-dup
soni

PII - har bor pattern boydcha maslak soni

Final - hujjat soni, belgi soni, taxminiy
token soni

data.yaml - fayl ichida data b-n qilingan
bilar haqida ma'lumot yoziladi

Coordinated Universal Time (UTC)

formerly

Greenwich Mean Time (GMT)

Sun

12:00

Number indicates standard time
in 2006 when it is 12 noon, UTC

South Georgia and the
South Sandwich Islands
(administered by U.K.
claimed by ARGENTINA)